

Edo Danilyan

PhD Researcher | Microbial Physiology | Bioinformatics | Data Science | Bioprocess

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PROFILE

PhD researcher in microbial physiology at the University of Vienna working across quantitative experiments, automated continuous cultivation, single-cell measurement, bioinformatics, and reproducible data analysis. Research background spans microbial growth and biomass, 16S rRNA amplicon analysis, functional interpretation of metagenome-assembled genomes (MAGs), microbial biotechnology, process optimization, environmental microbiology, geospatial analysis, and scientific visualization. Record includes 14 scholarly outputs since 2020, including two first-author papers, alongside open-source data projects and multi-institution scientific outreach.

RESEARCH AND PROFESSIONAL EXPERIENCE

University of Vienna - Centre for Microbiology and Environmental System Science | Vienna, Austria

PhD Researcher, FWF Cluster of Excellence MicroPlanet | Jul 2024 - Present

- Investigate how microbial growth, cell size, biomass, and metabolism respond to controlled nutrient limitation using automated chemostats, microbial cultivation, process monitoring, single-cell measurements, microscopy, and quantitative analysis.
- Redesigned an experimental workflow from ~12 L of medium for 1 data point/week to ~2 L for 7 data points/week, increasing throughput 7x while reducing medium use per result by ~98% and lowering consumable demand.
- Developed a low-perturbation sampling strategy for pulsed chemostats, enabling repeated sampling while keeping process disturbance within the magnitude of routine dilution cycles.
- Designed multi-condition continuous-cultivation experiments integrating dilution-rate control, process monitoring, sampling, LifeScale single-cell buoyant-mass measurements, and reproducible downstream analysis.
- Built automated R-based quality-control logic to flag process deviations, assess stability over time, and support data-driven decisions on sampling and downstream analysis.
- Diagnosed long-running automated bioreactor failures through structured root-cause analysis across fluidics, mixing, sampling, sensors, and culture conditions.
- Work on a microscopy-based cell-volume estimation pipeline to generate quantitative cell-volume estimates across growth conditions and compare them with complementary single-cell measurements.
- Developed and maintain the research-group website since 2025, translating research outputs into accessible digital content and managing ongoing technical updates.

Institut Teknologi Sepuluh Nopember | Surabaya, Indonesia

BSc & MSc Research - Microbial ecology, biotechnology, bioinformatics, and quantitative biology | 2018 - 2023

- Processed 16S rRNA amplicon datasets using mothur for demultiplexing and sequence quality control, DADA2 for denoising/sequence-variant processing, and R/tidyverse for downstream data handling, statistics, and visualization.
- Screened limestone-quarry microbial communities using 16S amplicon sequencing to identify candidate carbonatogenic bacteria, then isolated and selected *Lysinibacillus fusiformis* JH2 for microbial-geopolymer experiments.
- Combined microbial cultivation with XRD and SEM characterization in a first-author study that reported 17.5 MPa compressive strength at 7 days and up to a **166%** increase in strength.
- Performed functional analysis of MAGs across three research contexts - bachelor final project, master's thesis work, and a cross-university collaboration - to interpret microbial metabolic and ecological potential.
- Built phylogenetic analyses and publication-ready tree visualizations with ggtree and used ggplot2/tidyverse for microbial-community and functional-analysis results.

University of Szczecin | Szczecin, Poland

Exchange Researcher, Biology - Erasmus+ | Oct 2022 - Feb 2023

- Completed a semester-long research exchange with a GPA of 4.0/4.0 and contributed to a computational biology collaboration on microRNA as a melanoma biomarker.
- Co-authored a peer-reviewed narrative review in the International Journal of Molecular Sciences on diagnostic and therapeutic applications of microRNAs in cutaneous malignant melanoma.

Generasi Biologi Indonesia Foundation | Remote / Indonesia

Research Intern, Molecular Biology and Genetics | Oct 2021 - Feb 2022

- Conducted compound-target interaction studies using molecular docking and natural-product modeling, contributing computational evidence to molecular-biology research questions.

BRIN - National Research and Innovation Agency | Bogor, Indonesia

Student Research Intern, Bioprocess Laboratory | Jul 2021 - Sep 2021

- Evaluated nutrient-medium effects on *Coellastrella* sp. microalgal growth and produced reproducible R visualizations and statistical interpretation of growth responses.
- Translated the work into an innovation proposal recognized as a semi-finalist in the 2021 APEC YES Challenge.

Kumamoto University / Institut Teknologi Sepuluh Nopember | Remote

Collaborative Online International Learning | Oct 2020 - Nov 2020

- Co-developed a microbe-based waste-to-energy proposal for Surabaya and analyzed public waste data with statistical and geospatial tools in an international team.

National Yang Ming University | Remote

Pre-Internship Student - Drug discovery and computational molecular biology | Jan 2021 - Apr 2021

- Completed pre-internship projects focused on natural-product exploration, drug discovery, and computational molecular docking.

Dealls - Jobs & Mentoring | Online

Super Mentor | Jun 2024 - Present

- Provide one-to-one mentoring on research, bioinformatics, data science, graduate applications, career transitions, and academic development in 30-60 minute online sessions.

Ngelesin Global Edutech | Jakarta, Indonesia

Business Development Specialist | Jul 2022 - Apr 2023

- Supported business-development initiatives, growth planning, and partnership activities for an Indonesian education-technology platform.

TECHNICAL METHODS AND TOOLS

Microbial physiology & bioprocess: automated and pulsed chemostats; continuous cultivation; nutrient limitation; dilution-rate control; microbial growth experiments; OD-based growth analysis; sampling strategy; process stability; fluidics, mixing, sensors, and experimental troubleshooting.

Single-cell & imaging: LifeScale single-cell buoyant-mass measurements; biomass and cell-size analysis; microscopy-based cell-volume estimation; integration of complementary measurement streams.

Bioinformatics & computational biology: 16S rRNA amplicon workflows; mothur demultiplexing and QC; DADA2; microbial community analysis; MAG functional analysis; ggtree; Bioconductor; phylogenetic analysis; molecular docking; PyMOL.

Statistics & data science: R (tidyverse, dplyr, ggplot2); experimental design; hypothesis testing; statistical modelling; multivariate analysis; automated QC; anomaly/deviation detection; time-series analysis and forecasting training; LULC machine-learning implementation; Python (pandas, matplotlib, scikit-learn); SQL.

Geospatial & visualization: QGIS; sf/terra/raster workflows; geospatial Python/R; Leaflet; rayshader; D3; JavaScript; SVG; Shiny; Tableau; Plotly; flexdashboard; interactive maps and visual essays.

Reproducibility & development: Quarto; R Markdown; Git/GitHub; automated reporting; Hugo; VS Code; FAIR principle; structured data-quality checks and reusable analysis code.

SELECTED DATA, VISUALIZATION, AND OPEN COMPUTATIONAL PROJECTS

[The Pacific's Broken Ring](#) | R | D3 | Quarto | spatial analysis | 2026

Processed 358,673 global M4+ earthquakes across 26 years, defined a reproducible circum-Pacific working corridor, and explored event density, depth, magnitude-frequency behavior, temporal clustering, and boundary sensitivity.

[Living with a Changing Climate Across the Pacific](#) | R | D3 | JavaScript | Quarto | 2026

Integrated multiple public datasets spanning emissions, ocean temperature, sea level, cyclones, reef heat stress, rainfall, crop yield, and electricity. The analysis includes 508 South Pacific cyclone tracks and 450 rainfall-yield observations with documented limitations and a reproducible tidyverse data build.

[Bird Migration - How Three Gulls Found Their Own Way South](#) | R | geospatial analysis | D3 | SVG | 2026

Analyzed 61,920 GPS positions covering 259 days and 69,001 km for three tagged gulls; derived movement, timing, stopover, and route features, integrated historical weather, and explicitly handled tracking gaps longer than six hours.

[Living with Volcanoes](#) | R | D3 | rayshader | Quarto | 2025

Combined volcano and population data to examine human exposure around Holocene volcanoes on Java, including the estimate that about 60 million people live within 30 km of a Holocene volcano.

[Growing, but Not Yet Steady](#) | R | D3 | Quarto | microbiology | 2026

Built an experimental data story about when microbial populations reach effective steady state, connecting dilution, growth trajectories, and the limits of visual judgment from culture turbidity alone.

[My Year in Music - A Year in 22,197 Plays](#) | R | JavaScript | Quarto | personal data | 2026

Reconstructed a year of Spotify listening as a temporal data story, using listening history to examine routines, thesis pressure, travel, and changing patterns over time.

[One Soup, Many Islands](#) | D3 | JavaScript | Quarto | network analysis | 2026

Explored similarities among regional Indonesian soto recipes using ingredient relationships and interactive network storytelling.

[Spain's Power Grid](#) | R | Quarto | energy dashboard | 2026

Explored electricity generation, demand, weather, renewable share, and energy-mix change in Spain through an interactive analytical dashboard.

EDUCATION

University of Vienna | Vienna, Austria

PhD in Microbiology and Environmental Science | 2024 - Present | FWF Cluster of Excellence MicroPlanet: Microbiomes Drive Planetary Health

Institut Teknologi Sepuluh Nopember | Surabaya, Indonesia

MSc in Biology, GPA 4.0/4.0 | 2021 - 2023 | Fast Track and Fresh Graduate Scholarships; Most Outstanding Graduate

Institut Teknologi Sepuluh Nopember | Surabaya, Indonesia

BSc in Biology, GPA 3.79/4.0 | 2018 - 2022 | Bidikmisi Scholarship

University of Szczecin | Szczecin, Poland

Erasmus+ Exchange in Biology, GPA 4.0/4.0 | Oct 2022 - Feb 2023 | Erasmus+ Scholarship

Chulalongkorn University | Bangkok, Thailand

Exchange Studies in Biology, GPA 3.96/4.0 | Jan 2020 - May 2020 | ASEAN & NON-ASEAN Scholarship; special project in biological control

PUBLICATIONS

14 scholarly outputs listed on edodanilyan.com/publication/. First-author papers are marked below.

2026. Anjar Tri Wibowo, Arif Luqman, Fahmi Ikhlus Amalludin, Edo Danilyan, Yohanes Kartjito Putro, Elsalisa Ainur Rofiq, Almando Gerald, Yosephine Sri Wulan Manuhara, Hoang Dang Khoa Do, Minh Thiet Vu, Le Minh Bui, Wahyu Aristyaning Putri. [Metagenomic and Metabolomic Analysis Reveals Organ- and Population-Specific Interactions Between Bacterial Communities and Metabolites in Medicinal Plants *Tinospora cordifolia* and *Gynura procumbens*](#). *Advancements in Life Sciences*.
2025. Moch Ilham Abdillah, Aunurohim Aunurohim, Dian Saptarini, Miftakhul Sefti Raufanda, Edo Danilyan. [Characteristics of Microplastics and Comparison of Estimated Daily Intake Levels \(EDI\) in Consumption Shells in Kenjeran and Balekambang](#). *BIO Web of Conferences*.
2025. Nafilah Aulia Amanda, Aunurohim Aunurohim, Dian Saptarini, Miftakhul Sefti Raufanda, Edo Danilyan. [Bioaccumulation of Microplastic in Intertidal Gastropods from Southern Java, Indonesia: Abundance in Relation Feeding Guild](#). *BIO Web of Conferences*.
2025. Edo Danilyan, Januarti Jaya Ekaputri, Enny Zulaika, Puput Risdanareni, Didier Snoeck. [Utilization of soil microbial community to enhance mechanical properties of geopolymer paste for sustainable construction materials](#). *Case Studies in Construction Materials*. First author
2025. Aunurohim, Dian Saptarini, Abdus Junaedi, Shinta Dewi, Edo Danilyan, Hasan Adro'i, Prima Putra, Alfiah Hayati. [Characteristics of Microplastics in Water and Fish and Their Relationship with Migration from the East Coast of Surabaya, Indonesia](#). *Egyptian Journal of Aquatic Biology & Fisheries*.
2024. Aunurohim Aunurohim, Miftakhul Sefti Raufanda, Dian Saptarini, Farid Kamal Muzaki, Romanus Edy Prabowo, Martinus Surya Ari Pamungkas, Sobrian Cahya Perdana Putra, Ekawati Ekawati, Nur Maulidhatuffrida, Shinta Cahyaning Dewi, Edo Danilyan. [Microplastic-Contamination in The Flesh and Gastrointestinal Tract of Nile Tilapia \(*Oreochromis niloticus*\) reared in Floating Net Cages at Lake Ranu Grati, Pasuruan, East Java, Indonesia](#). *Environmental Advances*.
2024. Chantika Wijayanti, Nengah Dwianita Kuswytasari, Maya Shovitri, Edo Danilyan, Enny Zulaika. [Fungal diversity in the temple with an Internal Transcribed Spacer \(ITS\) molecular and morphological approach](#). *Biodiversitas Journal of Biological Diversity*.
2023. Nurul Fitria, Dewi Hidayati, Fredy Kurniawan, Arif Luqman, Noor Nailis Sa'adah, Edo Danilyan. [Acceptability Test of Modified Transglutaminase Gelatin from Striped Catfish \(*Pangasius hypophthalmus*\) Skin Based on Organoleptic and Toxicology](#). *Advancements in Life Sciences*.
2023. Edo Danilyan, Arif Luqman, Januarti Jaya Ekaputri, Enny Zulaika. [Exploring the Hydrocarbon Degradation Potential of Indigenous Bacteria using 16S Sequencing Approach](#). *International Journal of Research in Engineering, Science and Management*. First author
2023. Agata Poniewierska-Baran, Lukasz Zadroga, Edo Danilyan, Paulina Malkowska, Paulina Niedzwiedzka-Rystwej, Andrzej Pawlik. [MicroRNA as a Diagnostic Tool, Therapeutic Target and Potential Biomarker in Cutaneous Malignant Melanoma Detection - Narrative Review](#). *International Journal of Molecular Sciences*.
2022. Nur Hidayatul Alami, Afan Hamzah, Bieby Voijant Tangahu, Idaa Warmadewanti, Ary Bachtiar Krishna Putra, Adi Setyo Purnomo, Edo Danilyan, Hellen Melati Putri, Citra Nesa Aqila, Aulia An Nisaa Dewi, Ayudia Pratiwi, Salsabilla Khairunisa Putri, Arif Luqman. [Microbiome profile of soil and rhizosphere plants growing in traditional oil mining land in Wonocolo, Bojonegoro, Indonesia](#). *International Journal of Phytoremediation*.
2021. Iska Desmawati, Angga Gian, Rizki Widyaningsih, Zulfan Febriawan, Saniah Kusnatur, Ory Kurnia, Alwi Muhammad, Dimas Wildan, Edo Danilyan. [Dry Season Bryophytes In Cangar Hotspring-Batu, East Java, Indonesia](#). *Ecology, Environment and Conservation Paper*.
2020. N. Samboon, P. Ouying, W. Wannasrichan, N. Thoraneenitany, T. Achawong, K. Supatniyapong, K. Weerasopon, C. Khayunkan, Edo Danilyan, M. Hermanto, S. Kadkanla. [In Vitro Point-of-Care Diagnostic Assays for SARS-CoV-2 \(COVID-19\)](#). *Buddhachinaraj Medical Journal*.
2020. Arif Luqman, Nur Hidayatul Alami, Edo Danilyan, Bethari Auchenfloretta, Berendina Fionna Dama. [Bakteriologi Spesies Kosmopolit](#). *Biology Department, Institut Teknologi Sepuluh Nopember*. Book

TEACHING, MENTORING, AND SCIENTIFIC COMMUNICATION

- Secured funding and led a two-week MicroPlanet Global Ambassador outreach program across three educational institutions in Indonesia in 2026, combining microbial-growth lectures, open-source bioreactor demonstrations, R training, data storytelling, and graduate-study mentoring; the program reached more than 200 students and faculty members.
- Co-developed a hands-on Microbial Systematics course module covering microbial taxonomy, classification, and phylogenetic analysis in R.
- Mentor undergraduate and graduate students on research planning, bioinformatics, data analysis, graduate applications, and early-career decisions, including one-to-one mentoring through Dealls.
- Developed public R tutorials and open computational resources covering bacterial growth analysis, visualization, bioinformatics, and interactive biological data exploration.

SELECTED INVITED TALKS AND WORKSHOPS

- Apr 2026 - Storytelling with Data | Universitas Muhammadiyah Malang | faculty and researchers
- Apr 2026 - Why Do Cells Grow the Way They Do? Probing Microbial Growth Laws with an Automated Open-Source Bioreactor | Universitas Muhammadiyah Malang
- Apr 2026 - Preparing for Graduate Studies Overseas: Applications, Funding, and Mentorship | Institut Teknologi Sepuluh Nopember
- Apr 2026 - R for Biologists: Data Wrangling, Visualization, and Reproducible Analysis | Institut Teknologi Sepuluh Nopember
- Jan 2025 - Storytelling with Data | Bioinformatics Study Club, Institut Teknologi Sepuluh Nopember

AWARDS AND FUNDING

- MicroPlanet Global Ambassador Program funding - secured and led multi-institution scientific outreach in Indonesia (2026).
- Most Outstanding Graduate, Institut Teknologi Sepuluh Nopember - Master's Degree.
- Best Presenter, International Conference of Science and Technology (ICOST), 2023.
- Erasmus+ Scholarship, University of Szczecin exchange, 2022-2023.
- Top 10 Finalist, Battle of Minds Competition by British American Tobacco.

- 3rd Place, International Business Competition ICoBioS/OBC, 2022.
- Fresh Graduate Scholarship, Institut Teknologi Sepuluh Nopember.
- Fast Track Scholarship, Institut Teknologi Sepuluh Nopember.
- Semi-Finalist, APEC YES Challenge, 2021.
- Gold Medal and Best Impact Paper, Indonesia Inventor Days, 2020.
- Gold Medal and MIICA Special Award, World Invention Competition and Exhibition (IYSA).
- ASEAN & NON-ASEAN Scholarship, Chulalongkorn University.
- Bidikmisi Scholarship, undergraduate study.

SELECTED TRAINING AND CERTIFICATIONS

- 2024 - Python for Geospatial Raster Data, GeosoftwareID.
- 2024 - Land Cover Classification using Satellite Imagery with Different Machine Learning Algorithms.
- 2024 - Time Series Data Analysis and Visualization and Making Forecasts with R.
- 2022 - Sustainable Agriculture, National Dong Hwa University (Taiwan Ministry of Education-supported program).
- 2021 - Object-Oriented Programming in R, DataCamp.
- 2021 - Surveyor Training XI, Ecological Laboratory, Department of Biology, Institut Teknologi Sepuluh Nopember.
- 2021 - R Programming Fundamentals, edX.
- 2020 - Data Science: R Basics, edX.

LEADERSHIP AND ACTIVITIES

Kelompok Studi Burung Liar Pecuk | Surabaya, Indonesia

Staff, Multimedia Division | Sep 2019 - Jan 2021

- Contributed wildlife photography to bird-identification archives and public outreach and supported content development for a monthly digital magazine focused on avian ecology.

Chulalongkorn University Debating Society | Bangkok, Thailand

Member | Jan 2020 - May 2020

- Participated in weekly competitive-debate training and was selected for the university national-championship preparation team.

LANGUAGES

Bahasa Indonesia (native) | English (full professional proficiency) | German (basic) | Japanese (basic)